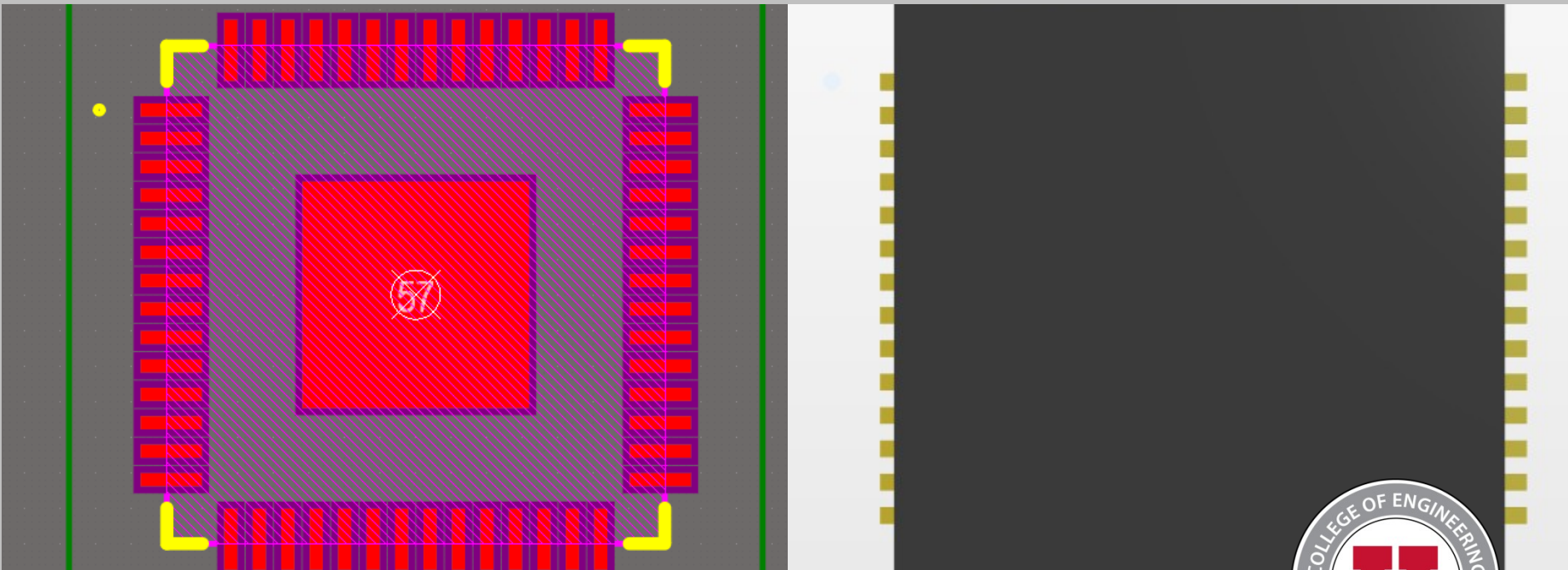


IEEE PCB Design Workshop: (Week 04) Footprints and PCB Layout



Hosted By: Adrian Sucahyo and IEEE at the University of Utah

Workshop Outline

Tentatively scheduled for 6-7 weeks

- Sept. 15 – Introduction to Schematics (WEB L102)
- Sept. 22 – Schematics Continued (WEB L112)
- Sept. 29 – Introduction to PCB Layout (WEB L102)
- **Oct. 6 – Layout Continued (WEB L102)**
- Oct. 8 – PCBs DUE!
- **** FALL BREAK ****
- TBD – Soldering Week 1
- TBD – Soldering Week 2
- TBD – Soldering Week 3 (Optional)

Announcements

- Alternative Projects
 - We will be able to get alternative project boards manufactured if submitted by the deadline.
 - Limited to the 10 cm x 10 cm dimensions outlined by JLCPCB.
 - Talk or email me if you have any questions!
- PCB Submission Deadlines
 - For both workshop and custom PCBs, the deadline for submission is **Oct 8th**

Want more experience?

- Consider joining the FSAE tractive team!
 - The Tractive Team is currently looking for students to assist with designing and assembling the electrical system for an electric formula-style race car!
 - No experience required!

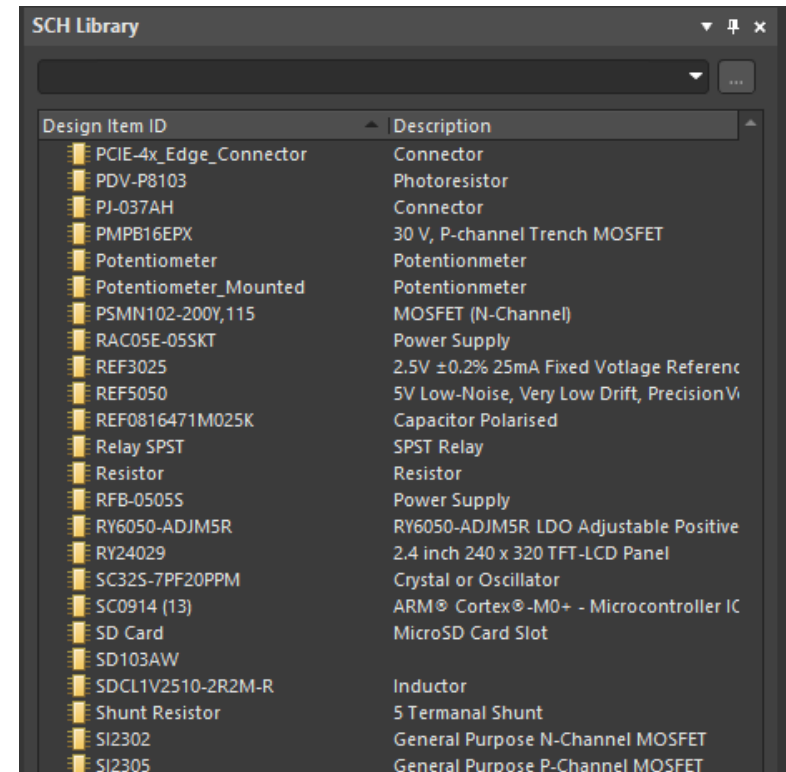


U of U FSAE Discord Link



Component Libraries Refresher

- Libraries are collections of components and footprints
 - All components for a project must be derived from a library
 - Projects may reference multiple different component libraries
 - Relatively consistent across platforms
- Typically, there are symbol and footprint libraries
 - Tightly coupled together



The screenshot shows a software window titled "SCH Library". It contains a table with two columns: "Design Item ID" and "Description". The table lists various electronic components and their specifications.

Design Item ID	Description
PCIE-4x_Edge_Connector	Connector
PDV-P8103	Photoresistor
PJ-037AH	Connector
PMPB16EPX	30 V, P-channel Trench MOSFET
Potentiometer	Potentiometer
Potentiometer_Mounted	Potentiometer
PSMN102-200Y,115	MOSFET (N-Channel)
RAC05E-05SKT	Power Supply
REF3025	2.5V $\pm 0.2\%$ 25mA Fixed Voltage Reference
REF5050	5V Low-Noise, Very Low Drift, Precision Voltage Reference
REF0816471M025K	Capacitor Polarised
Relay SPST	SPST Relay
Resistor	Resistor
RFB-05055	Power Supply
RY6050-ADJM5R	RY6050-ADJM5R LDO Adjustable Positive
RY24029	2.4 inch 240 x 320 TFT-LCD Panel
SC325-7PF20PPM	Crystal or Oscillator
SC0914 (13)	ARM [®] Cortex [®] -M0+ - Microcontroller IC
SD Card	MicroSD Card Slot
SD103AW	
SDCL1V2510-2R2M-R	Inductor
Shunt Resistor	5 Terminal Shunt
SI2302	General Purpose N-Channel MOSFET
SI2305	General Purpose P-Channel MOSFET

Standard vs. Non-Standard Footprints

- Many components are in “standard” packages
 - Key parameters are standardized across manufacturers
- Certain components are in “non-standard” footprints due to custom features or other restrictions



<https://www.indiamart.com/proddetail/smd-electronic-component-2856299588912.html>

Resistors and Capacitors

- Many components are in “standard” packages
 - Key parameters are standardized across manufacturers
- Certain components are in “non-standard” footprints due to custom features or other restrictions

Imperial Size Code Metric Size Code

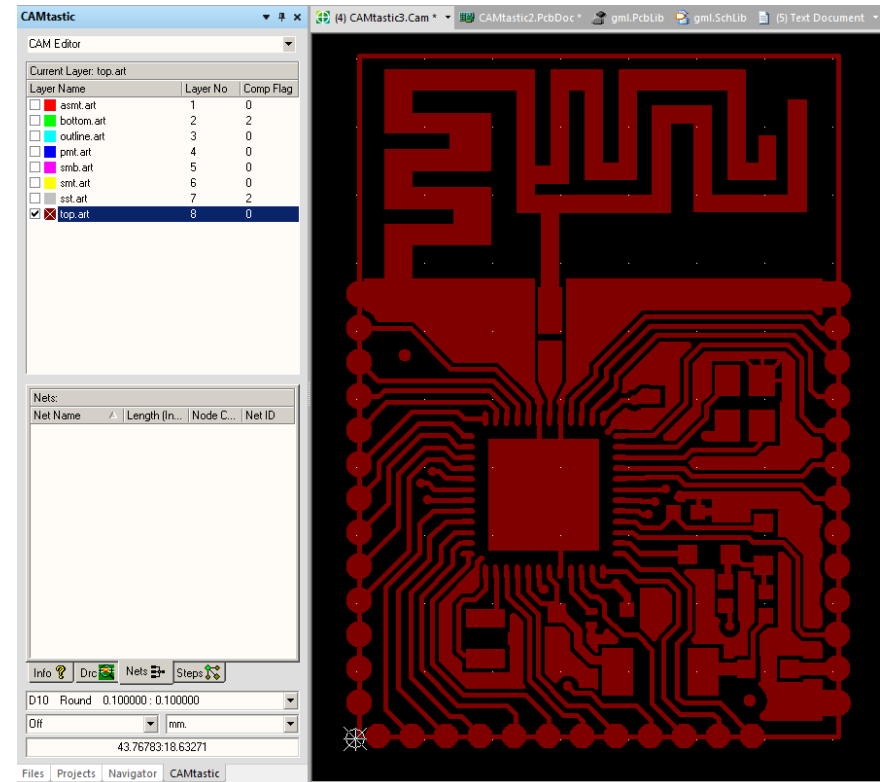
	01005	0402	
	0201	0603	
	0402	1005	
	0603	1608	
	0805	2012	1x1cm
	1008	2520	
	1206	3216	
	1210	3225	
	1806	4516	1x1cm
	1812	4532	
Size	2010	5025	Size
Reference	2512	6332	Reference

Component Sizes (common in bold)

https://developer.wildernesslabs.co/Hardware/Reference/Components/Packages_and_Sizes/

Gerber Files

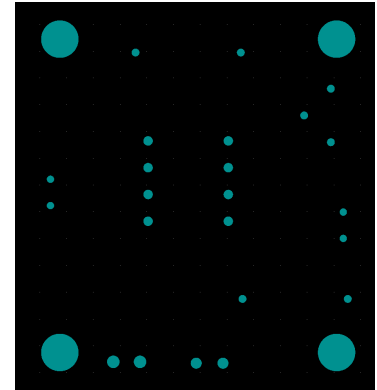
- Gerbers (or Gerber Files) are the files sent to the manufacturer that will be used to make the boards
- Engineers at the fab review files for manufacturability and then send it off to production.
- Gerber files can be generated from EDA software but needs to be reviewed.



<https://www.onethesis.com/gerbers-to-footprint/>

NC Drill Files

- NC Drill Files contain the information for the drill holes of the PCB.
 - These are generated separately from the Gerber Files



----- NCDrill File Report For: Workshop Project.PcbDoc 10/29/2024 1:09:38 AM -----						
Layer Pair : Top Layer to Bottom Layer ASCII RoundHoles File : Workshop Project.TXT						
Tool	Hole Size	Hole Tolerance	Hole Type	Hole Count	Plated	Tool Travel

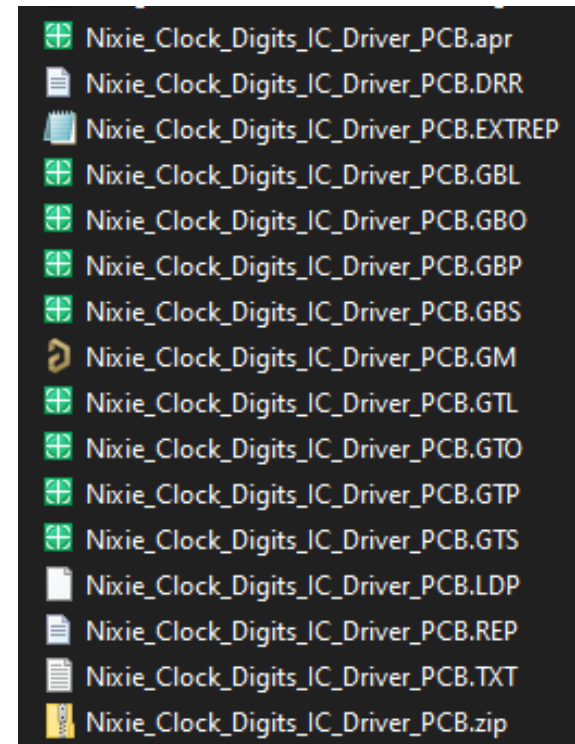
T1	28mil (0.7mm)		Round	4	PTH	1.29inch (32.82mm)
T2	30mil (0.75mm)		Round	7	PTH	2.02inch (51.35mm)
T3	35mil (0.9mm)		Round	8	PTH	0.90inch (22.86mm)
T4	41mil (1.05mm)		Round	2	PTH	0.10inch (2.54mm)
T5	47mil (1.19mm)		Round	2	PTH	0.10inch (2.54mm)
T6	140mil (3.556mm)		Round	4	PTH	3.24inch (82.30mm)

Totals				27		

Total Processing Time (hh:mm:ss) : 00:00:00						

Gerber File and Drill File Types

- Each file represents a different layer or operation type.
- Common files:
 - .gto, .gbo = top/bottom silkscreen
 - .gts, .gbs = top/bottom solder mask
 - .gtp, .gbp = top/bottom solder paste
 - .gtl, .gbl = top/bottom copper
 - .gm = board outline
 - .drl / .txt = CNC drill



Questions?

Questions?

Download Today's Project Files

Navigate to the workshop GitHub and
download today's files

<https://github.com/IEEE-U-of-U/IEEE-PCB-Workshop-Fall-2026>